

Homework 10

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Section 003
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Collaboration Statement

I worked on this homework independently. No outside collaborators or resources were used.

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Problem 1 (*Ross Chapter 3 18.5 (a)*)

Let f and g be continuous functions in $[a, b]$ such that $f(a) \geq g(a)$ and $f(b) \leq g(b)$. Prove $f(x_0) = g(x_0)$ for at least one x_0 in $[a, b]$.

Discussion.

Proof.

□

Problem 2 (*Ross Chapter 3 18.10*)

Suppose f is continuous on $[0, 2]$ and $f(0) = f(2)$. Prove there exist x, y in $[0, 2]$ such that $|y - x| = 1$ and $f(x) = f(y)$. *Hint:* Consider $g(x) = f(x + 1) - f(x)$ on $[0, 1]$.

Discussion.

Proof.

□

Problem 3 (*Ross Chapter 3 19.6 (b)*)

(From Part (a):) Let $f(x) = \sqrt{x}$ for $x \geq 0$. Show f' is unbounded on $(0, 1]$ but f is nevertheless uniformly continuous on $(0, 1]$ Compare with Theorem 19.6.

Show that f is uniformly continuous on $[1, \infty)$.

Discussion.

Proof.

□

Problem 4 (*Ross Chapter 3 19.7*)

(a) Let f be a continuous function on $[0, \infty)$. Prove that if f is uniformly continuous on $[k, \infty)$ for some k , then f is uniformly continuous on $[0, \infty)$.

(b) Use (a) and Exercise 19.6(b) to show that $f(x) = \sqrt{x}$ is uniformly continuous on $[0, \infty)$.

Discussion.

Proof.

□

Problem 5 (*Uniform Continuity of f*)

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$. Prove that if there exists a real number $M \geq 0$ such that $|f(x) - f(y)| \leq M|x - y|$ for all $x, y \in \mathbb{R}$, then f is uniformly continuous on $\mathbb{R}$.

Discussion.

Proof.

□

Problem 6 (*Differentiability of f*)

Suppose that $0 \in (a, b)$ and $f(x) = xg(x)$ for some function $g : (a, b) \rightarrow \mathbb{R}$ which is continuous at $x = 0$. Prove that f is differentiable at $x = 0$, and find $f'(0)$ in terms of g .

Hint: Note that you cannot use the product rule, since $g(x)$ is not guaranteed to be differentiable at $x = 0$

Discussion.

Proof.

□